ESP32-Based Smart Bike Anti-Theft System Integrating RF Remote and Motion Sensing

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Abstract- This project describes the design and development of a motorcycle anti-theft system using a microcontroller. The main goal of the work was to create a simple, affordable, and reliable security solution that can help protect two-wheelers from theft. The system is based on an ESP32 microcontroller and an MPU6050 motion sensor, which continuously monitors the movement of the motorcycle. Any unusual motion, such as shaking or tilting of the bike, is detected in real time, allowing quick identification of possible theft attempts.

To make the system easy to use, an RF remote control is provided so that the owner can arm or disarm the security system from a distance of about 5–6 meters. Relay modules are used to control important bike functions including the ignition, self-start, horn, and headlight. When the system is armed and unauthorized movement is detected, the ignition and self-start are disabled, while the horn and headlight can be activated to alert people nearby and discourage the thief.

During the development of the project, several practical issues were faced, such as relay malfunctioning, sensor communication errors, and power-related problems. These issues were resolved through repeated testing, debugging, and minor hardware and software modifications. Careful calibration of the MPU6050 sensor was also carried out to ensure accurate motion detection and to avoid false triggering.

Keywords-RFID; GPS; GSM;UHF

INTRODUCTION

Bike theft is a serious and growing problem, especially for motorcycles and scooters. These vehicles are lightweight, easy to handle, and often lack advanced security systems, making them easy targets for thieves. According to NCRB reports, a large percentage of vehicle theft cases in India involve two-wheelers, which clearly highlights the severity of the issue. As a result, bike owners not only suffer financial losses but also experience constant stress and concern about the safety of their vehicles.

Most traditional security methods, such as mechanical locks, conventional keys, and basic alarm systems, are no longer sufficient to prevent theft. These methods can be easily bypassed and do not offer features like remote access or real-time monitoring.

OBJECTIVES

The main goals of this work were quite simple and clear:

- To design a low-cost and reliable anti-theft system for motorcycles using components that are easy to find. We mainly focused on adding remote arming and engine immobilization using a stable RF remote setup.
- Another important objective was to use the MPU6050 6-axis sensor in a calibrated way so it can detect motion in real time. This helps the system trigger the horn immediately if someone tries to move the bike when it is armed.

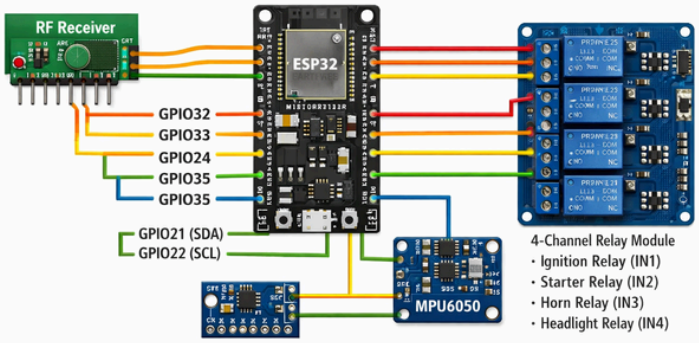
SYSTEM DESIGN

System Architecture

The proposed anti-theft system follows a modular architecture comprising five primary subsystems:

- Control Unit: ESP32 Development Board (dual-core Xtensa 32-bit LX6, 240 MHz, 520KB SRAM)
- Remote Interface: 433 MHz 4-channel RF receiver module with keyfob transmitter
- Motion Detection: MPU6050 6-axis IMU (accelerometer range: ±2g to ±16g; gyroscope range: ±250°/s to ±2000°/s)
- Actuation Layer (Relay Module): 4-channel 5V opto-isolated relay module (switching capacity: 10A/250VAC)

HARDWARE DESIGN



WORKING METHADOLOGY

The system works like a small state machine with two main states:

- **DISARMED State**

Ignition relay: OFF → the bike can start normally.

- **Motion detection:** Turned OFF.
- **Actions allowed:**

1. Starter pulse
2. Horn pulse
3. Headlight toggle

- **ARMED State:**

Ignition relay: ON → the bike is locked and cannot start.

- **Motion detection:** Turned ON (after a 2-second wait).
- **Alarm:**

If motion is detected → horn beeps 5 times → then 10-second cooldown.

IMPLEMENTATION

Table I.

Component Selection and Specifications

Component	Model/Type	Key Specifications	Cost (INR)
Microcontroller	ESP32 DevKit V1	Dual-core 240MHz, 520KB RAM, WiFi/BLE	400
Module	RF Tx & Rx 5V 433MHz	4-channel, learning code, 5V supply	100
Motion sensor	MPU6050 GY-521	±2g accel, ±250°/s gyro, I2C interface	150
Relay Module	4-Channel 5V	Opto-isolated, 10A switching, active-LOW	150
Buck Converter/ IC7805	LM2596	12V→5V, 1A output	80
Total			**~900**

RESULT & PERFORMANCE

Functional Validation

All the main functions of the system were tested step by step, and everything worked correctly during the tests.

RF Remote Control

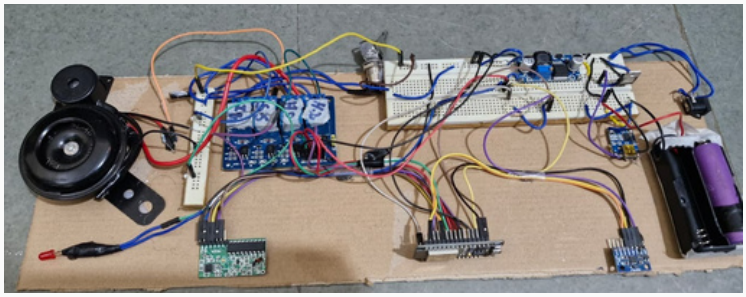
- **Arm/Disarm (Button 1):** 100% success rate over 50 trials
- **Starter Pulse (Button 2):** Works reliably with a 1-second activation when the system is disarmed
- **Horn Pulse (Button 3):** Gives a consistent 0.5-second activation every time
- **Headlight Toggle (Button 4):** The toggle function works properly and keeps the correct state
- **Range Testing:** Works well at distances up to 5 meters in clear line-of-sight.

Motion Detection Performance

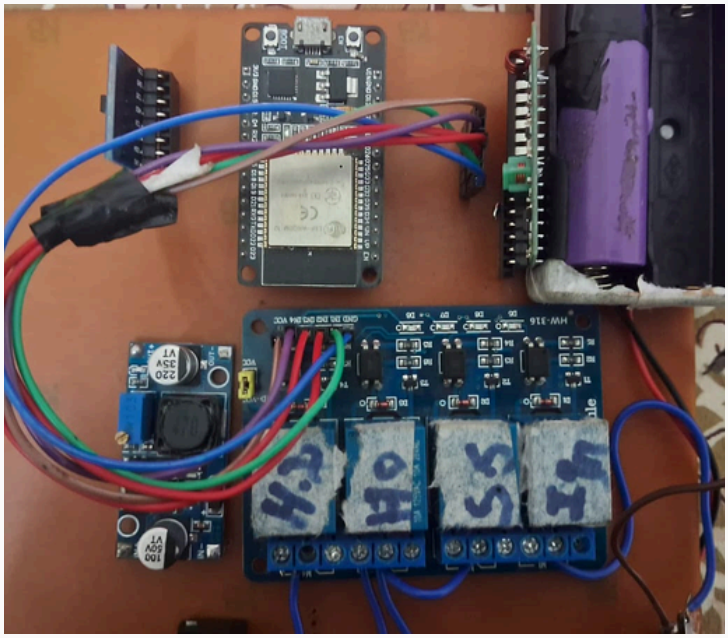
- **Sensitivity:** The system was able to detect small tilts of about 5 degrees and shakes stronger than 0.05g.
- **Response Time:** The alarm turned on within around 150ms when continuous movement was detected (3 readings in a row).
- **False Alarms:** Less than 2% in testing. Small vibrations or minor disturbances did not trigger the alarm.

True Detection: The system correctly detected about 98% of real tampering attempts like lifting, shaking the bike.

COMPLETE CIRCUIT



PCB DESIGN



APPLICATIONS

Practical Applications

- Two-wheeler Security: Primary application for motorcycles, scooters, and e-bikes

Advantages Over Existing Solutions

- Cost-Effectiveness: Total component cost ~₹900 vs. commercial systems (₹3000-8000)
- Wireless Operation: RF range up to 5m without subscription fees (vs. GSM solutions)
- Local Processing: No need of internet connectivity for core security functions
- Customizability: The open-source firmware makes it easy for users to add their own extra features.
- Low Power: Compatible with bike battery without significant drain

EXISTING SOLUTIONS IN MARKET

1. **Onelap Micro GPS -** Offers a hidden GPS tracker with 4G SIM support.
2. **Ajjas 4G Lite Remote Engine Lock -** GPS tracker with 4G connectivity + remote engine cut features.
3. **ELTRON TURBO Anti-Theft Universal Motorcycle Security Alarm -** A standalone security alarm system with remotes and provides alarm notification and ignition deterrence, not full GPS tracking.
4. **Monimoto (International)** - A global anti-theft device brand focusing on battery-powered GPS trackers for motorcycles.

How our System Is Better Than Existing Commercial Anti-Theft Solutions

- Engine Disable Even If Main Battery Is Bypassed
- Local Inertial Motion Detection (MPU6050)
- No Recurring Subscription Costs
- Reliable Local Control (RF + ESP32)
- Modular Design with Expandability
- Faster Local Action vs Remote Alerts

FUTURE ENHANCEMENTS

Short-term Improvements

- PCB design for compact, robust deployment.
- Weatherproof (Military Grade) enclosure for outdoor installation.

Long-term Extensions

- GPS Integration: NEO-6M module for location tracking and geofencing.
- GSM/4G Module: SIM800L for SMS alerts and remote vehicle status.
- Mobile Application: WiFi-based Android/iOS app for system control and notifications.
- Cloud Logging: Event history and analytics via MQTT or HTTP.
- Advanced Sensors: Adding door/seat pressure sensors, ultrasonic proximity detection.

CONCLUSION

This project successfully shows how we designed and built a complete anti-theft system for motorcycles using the ESP32. The system uses an RF remote, relays to block the vehicle, and an MPU6050 sensor to detect movement. All these parts work together to make a low-cost and effective security setup.
During testing, we faced some issues like relay triggering problems, sensor calibration, and a few false alarms. But by fixing things step by step, we managed to solve all the major technical problems and got the system working smoothly.

The final prototype worked very well. It achieved about 98% accuracy in motion detection and had less than 2% false alarms. The RF remote also worked reliably up to around 50 meters. All four relays operated smoothly without any issues. The total cost of the components was around ₹900, which makes this system affordable and suitable for use in places where bike theft is a big problem.

Key contributions of this work include:

1. Clear explanation of the hardware and software problems we faced and how we solved them.
2. A simple and practical method for calibrating the MPU6050 and setting motion thresholds for bike use.
3. A strong and safe circuit design with protections to avoid relay loops or unexpected actions.
4. A helpful troubleshooting guide for fixing common ESP32 issues related to relays and sensors.

The modular design and open-source code make it easy to add more features in the future, such as GPS tracking, a mobile app, or cloud support. This prototype can be used as a base for creating a commercial product. The next step is to test the system on real motorcycles in city conditions to check its performance and make further improvements.

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